

# AI-Assisted Lecture Analysis and Presentation Generation Report

## Introduction

The objective of this task was to analyze an orientation lecture video using Artificial Intelligence and generate a professional presentation covering the lecture topics. Since the lecture transcript was not readily available, an AI-assisted workflow was used to extract, understand, organize, and transform the lecture content into presentation slides. Multiple prompts were tested and refined throughout the process to achieve the desired outcome.

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## AI Tools and Models Used

The following AI tools were used during the task:

### 1. ChatGPT

Used for:

- Prompt engineering assistance
- Creating structured instructions
- Refining prompts
- Developing reporting documentation

### 2. Google Gemini

Used for:

- Processing the lecture video
- Understanding lecture content
- Extracting key concepts and learning points
- Generating presentation slides

The combination of ChatGPT and Gemini helped leverage the strengths of both tools to achieve a better final result.

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## Prompts Used During the Process

### Initial Prompt

The first prompt focused on understanding the lecture content:

"Analyze the provided lecture video and summarize the main topics, key concepts, learning objectives, examples, and important takeaways discussed by the instructor."

## Presentation Generation Prompt

A more detailed prompt was later used:

"You are an expert instructional designer, university lecturer, presentation creator, and note-taking assistant. Analyze the lecture thoroughly and generate detailed lecture notes, presentation slides, speaker notes, key takeaways, and important questions based on the lecture content."

This prompt included:

- Learning objectives
  - Detailed notes
  - Slide generation requirements
  - Speaker notes
  - Key takeaways
  - Final summary requirements
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## Which Prompts Worked Effectively

The most effective prompt was the detailed instructional-design prompt because it:

- Clearly defined the AI's role
- Specified the expected output format
- Broke the task into multiple steps
- Requested both analysis and presentation creation
- Included speaker notes and visual suggestions

The structured prompt produced significantly more organized and presentation-ready content than simple prompts.

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## Which Prompts Did Not Work Well

Some early prompts produced limited results.

Examples:

"Summarize this lecture."

"Create slides from this video."

These prompts were too broad and lacked:

- Output structure
- Formatting requirements
- Detail expectations
- Context regarding presentation quality

As a result, the responses were often brief and incomplete.

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## How the Prompts Were Refined

The prompting process followed an iterative improvement approach.

### First Iteration

A simple summary prompt was used.

Result:

- Basic summary
- Missing presentation structure
- Limited detail

### Second Iteration

Additional instructions were added.

Result:

- Better topic coverage
- Improved organization
- More useful notes

### Final Iteration

A comprehensive prompt was created that included:

- Role assignment
- Step-by-step instructions
- Lecture analysis requirements
- Presentation slide requirements
- Speaker notes
- Visual recommendations
- Key takeaways
- Questions and summary

Result:

- Detailed lecture understanding
- Professional presentation structure
- High-quality educational content
- Presentation-ready output

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## Prompting Techniques That Helped the Most

Several prompt engineering techniques improved the quality of results.

## **1. Role Prompting**

Assigning the AI the role of:

"Expert instructional designer, university lecturer, presentation creator, and note-taking assistant"

helped generate more professional educational content.

## **2. Structured Prompting**

Breaking the task into multiple clearly defined steps improved output quality significantly.

## **3. Output Formatting Instructions**

Specifying exactly what should be included in each slide reduced ambiguity and improved consistency.

## **4. Task Decomposition**

Instead of requesting everything at once, the process divided the task into:

- Lecture analysis
- Note creation
- Presentation development
- Key takeaway extraction

This produced more comprehensive results.

## **5. Iterative Refinement**

Each new prompt improved upon the weaknesses of the previous prompt until the desired outcome was achieved.

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## **Complete Process Followed**

### **Step 1**

Obtained the lecture video link.

### **Step 2**

Attempted to retrieve the lecture transcript.

### **Step 3**

When transcript extraction became difficult, alternative AI-assisted methods were explored.

## **Step 4**

Used ChatGPT to design a comprehensive prompt capable of analyzing lecture content and generating presentation materials.

## **Step 5**

Provided the improved prompt to Google Gemini.

## **Step 6**

Gemini analyzed the lecture content and generated:

- Lecture overview
- Key concepts
- Detailed notes
- Presentation slides
- Speaker notes
- Learning takeaways

## **Step 7**

Reviewed the generated output and evaluated its completeness.

## **Step 8**

Refined presentation structure and organized the content into a professional format suitable for academic submission.

## **Step 9**

Prepared the final presentation and documentation.

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## **Conclusion**

This task demonstrated the importance of effective prompt engineering when working with AI systems. Simple prompts produced limited results, while detailed and structured prompts generated significantly better outputs. By combining ChatGPT for prompt design and Gemini for lecture analysis, it was possible to successfully understand the lecture, extract key information, and create a professional presentation. The experience highlighted how role prompting, structured prompting, task decomposition, and iterative refinement can dramatically improve AI-generated results.